

Interview Practice Platform

AI Resume Matching & Feedback

Database Systems Engineering and Distributed Backend Development (25CS1302E)

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Frontend Design

Key screens in the candidate-facing React application.

Home Page

Landing page introducing the platform and entry point to upload a resume.

Login / Registration

Candidate authentication — sign up, log in, and session handling.

Forms

Resume upload and job-preference forms feeding the matching engine.

Dashboard

Match results, mock interview access and feedback history in one place.

Frontend Implementation — Screenshots

Interview Practice

Sign in to continue your prep.

Email

Password

→] Log in

or

Continue as guest

This is a demo login — any email/password works.

Role-Based Practice

Upload your resume

We'll match you to roles based on your skills and experience.



Drag and drop your resume, or click to browse

PDF or Word, up to 5MB

or

Paste your resume text instead

Find matching roles

[Use sample resume](#) · [Download it to upload manually](#)

Welcome back, Rasagnya

Here's how your interview prep is going.

3-day streak

72%

Interview readiness
Based on your last 14 sessions

Choose how you want to practice



Role-Based Practice

Upload your resume and get matched with roles, then practice questions tailored to that job.



Company-Based Practice

Pick a company and rehearse the rounds and question styles they're known for.



Coding / Technical Round

Solve timed problems in a mock editor and see how you'd score on a technical round.



HR / Behavioral Round

Practice the behavioral questions every interview opens with — and close with confidence.

 **Recommended for you**
Your coding round scores dipped this week. [Practice a coding round](#)

Skill breakdown

Averaged across your recent sessions



Your progress

Weekly Monthly

Sessions: 10

Avg score: 73%

Most practiced: Role-Based

Improvement: +8%

Score trend



Sessions by mode

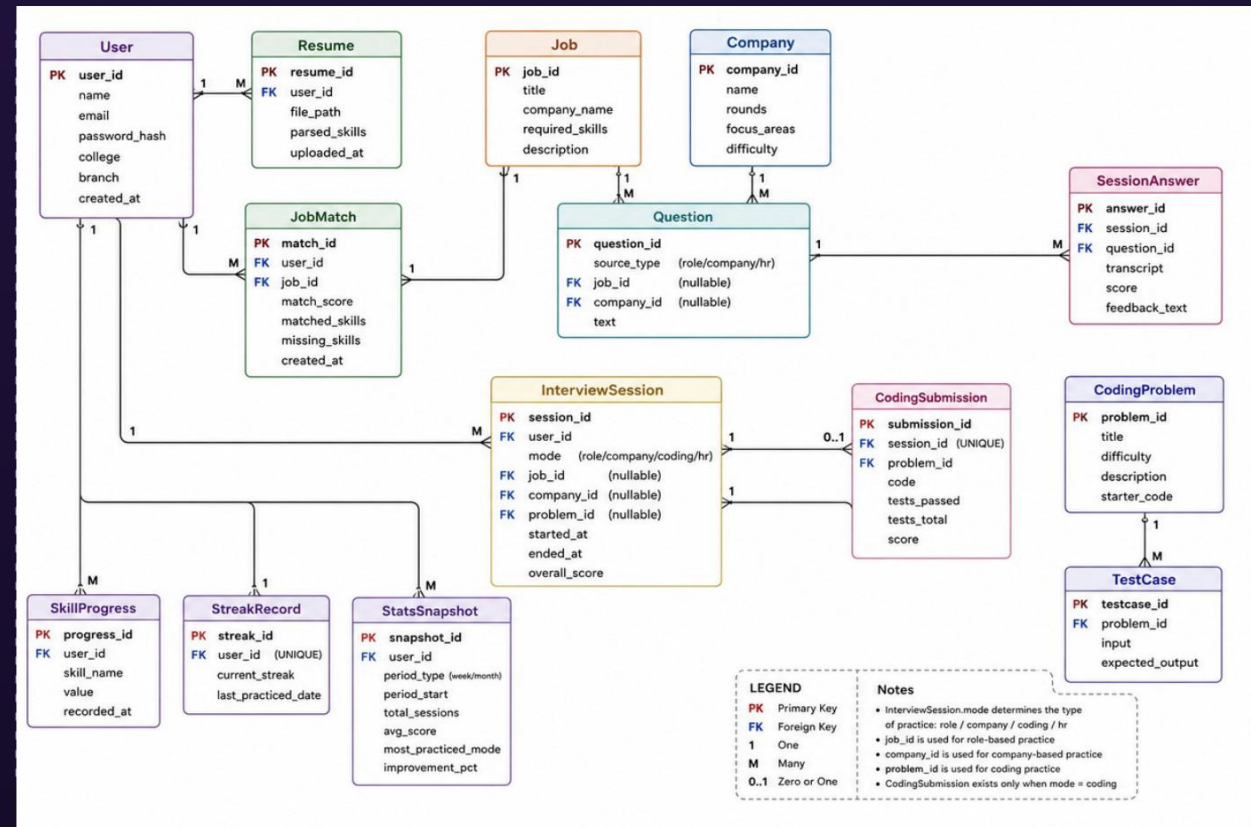


Database Design & ER Diagram

MongoDB collections for candidates, resumes, jobs, sessions and feedback, and how they relate.

Design Principles

- Flexible, document-based schema to suit variable-length transcripts
- Collections: Users, Resumes, Jobs, InterviewSessions, Feedback
- References between collections via ObjectId rather than deep nesting
- Indexes on candidate ID and session ID for fast dashboard lookups



Database Tables / Schema & Sample Records

Schema Overview

Users name, email, passwordHash, createdAt

Resumes userId, parsedSkills[], rawText, uploadedAt

Jobs title, description, requiredSkills[], embedding

InterviewSessions userId, jobId, questions[], transcript, status

Feedback sessionId, contentScore, deliveryScore, comments

InterviewIQ — Sample Database Records

Users

user_id	name	college	branch
1	Niyathi	KL University	IT
2	Rahul Sharma	ABC Engineering	CSE

Jobs

job_id	title	company_name
1	Data Analyst	Microsoft
2	Software Engineer	Google
3	BI Analyst	Amazon

InterviewSessions

session_id	user_id	mode	score
1	1	role	82.50
2	1	coding	90.00

Questions

question_id	source_type	question_text
1	role	Explain INNER JOIN vs
2	role	What is data normaliza
3	company	Why do you want to w
4	hr	Tell me about yourself.

Sample records from the InterviewIQ MySQL database

Current Progress

Resume parsing service

[In progress]

Job matching engine

[In progress]

Question generation (LLM)

[Not started]

Live answer capture (audio/video)

[Not started]

Speech-to-text pipeline

[Not started]

Feedback dashboard (frontend)

[Not started]

Update the status labels above with the team's actual progress before Review 2.

Challenges / Issues Faced

Open technical questions that come with this three-service, free-tier architecture.



Inconsistent resume formats

spaCy-based parsing works well on clean, single-column resumes but accuracy drops on tables, multi-column layouts and graphic-heavy templates — pushing us toward extra pre-processing and fallback rules rather than parsing alone.



Latency across three services

A single mock-interview turn chains Node → Flask → an LLM call (and Whisper for transcription). Keeping that feel responsive in a live, in-browser session means careful async handling, not just faster individual calls.



Free-tier rate limits

Groq/Gemini and MongoDB Atlas free tiers cap request volume and concurrency, so testing with multiple simultaneous mock sessions needs request queuing and graceful backoff rather than firing calls directly.



Cross-browser audio/video capture

getUserMedia and MediaRecorder support different codecs and permission flows across browsers, so a recording flow that works in one browser needs explicit fallback handling to stay consistent in others.

Future Work

● Body Language Analysis

MediaPipe-based heuristics for eye contact, posture and fidgeting during mock interviews.

● Recruiter Dashboard

A recruiter-facing view into candidate readiness and match pipelines.

● Multi-Language Interviews

Support mock interviews and feedback in regional languages.

● Voice Modulation Scoring

librosa-based pitch, pace and volume analysis for a fuller picture of delivery.

● Live Job Board Integration

Pull real postings via LinkedIn/Indeed APIs instead of a static seed dataset.

● Peer Mock Scheduling

Let candidates practice live with peers, with AI scoring both sides.

Conclusion

- The Interview Practice Platform brings resume-to-job matching and mock-interview feedback into a single workflow — something no single tool surveyed in the literature review does end to end.
- It replaces keyword-filtered resume screening with embedding-based matching (Sentence-BERT-style similarity), so candidates see roles that genuinely fit their skills, not just their keywords.
- It replaces generic question banks and delivery-only coaching with mock interviews generated from the matched job and the candidate's own resume gaps, scored on both content and delivery.
- Every component — spaCy, sentence-transformers, faster-whisper, Groq/Gemini, MongoDB Atlas — runs on free-tier or open-source infrastructure, so the platform stays accessible to the candidates who need it most.
- [Close with the team's specific next milestone toward the final review]



Thank You